

--	--	--	--	--	--	--	--

B.Tech. Degree IV Semester Examination April 2016

CS/IT 1405 DATA STRUCTURES AND ALGORITHMS (2012 Scheme)

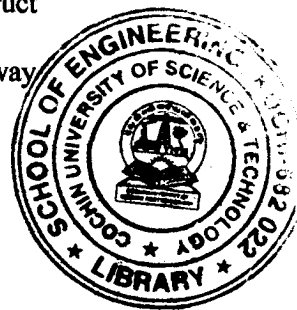
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) What is sparse matrix? Give its representation.
- (b) Write an algorithm to perform insertion sort.
- (c) Given two singly linked lists L1 and L2. Write an algorithm to merge two linked lists L1 and L2 into a single linked list.
- (d) Explain deque.
- (e) Is expression tree a binary tree? What is expression tree? Construct expression tree for the prefix expression $*+ab-cd$.
- (f) What are the needs for creating threaded binary tree? What is two-way threading?
- (g) Give array adjacency list representation of graph in memory.
- (h) Explain Kruskal's algorithm to construct minimum cost spanning tree.



PART B

(4 × 15 = 60)

- II. (a) Explain merge sort algorithm with example. Mention its complexity. (10)
 - (b) Write an algorithm to delete an element from an array. (5)
- OR**
- III. (a) Write short notes on: (8)
 - (i) Hash table.
 - (ii) Hash function
 - (iii) Collision.
 - (iv) Linear probing.
 - (b) What is max heap? Explain heap sort algorithm with example. (7)
- IV. Write an algorithm to translate an infix expression to postfix. Convert the infix expression $(A+B) \uparrow C - (D * E) / F$ to postfix. Note that $\uparrow$ is exponentiation operator. (15)
- OR**
- V. (a) What are the advantages of circular queue over queue? Write an algorithm to insert an element into a circular queue and delete an element from a circular queue. (10)
 - (b) Differentiate singly linked list and doubly linked list. (5)

(P.T.O.)

- VI. (a) Develop a non-recursive algorithm to perform in order traversal of binary tree. (9)
(b) Explain AVL tree. (6)
- OR**
- VII. What is binary search tree? How can insertion and deletion be done in a binary search tree? (15)
- VIII. Explain graph traversal methods. (15)
- OR**
- IX. (a) Write short notes on: (5 × 2 = 10)
(i) Undirected graph.
(ii) Self edge.
(iii) Spanning tree.
(iv) Degree.
(v) B-tree.
(b) Explain Dijkstra's algorithm, with an example. (5)

